



A Systematic Review of the use of Machine Learning in the Prediction of House Pricing

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Abstract

The real estate market involves large sums of money and is linked to the entire supply chain of the construction industry. The lack of standardization in property prices and the high fluctuations in the sector present opportunities for the use of machine learning (ML) in this type of problem. This systematic literature review analyzed research trends and identified research gaps in the field of ML applied to real state pricing. The results elucidated that the most frequently used models were Random Forest, Gradient Boosting Machine and XGBoost, Linear Regression, and Artificial Neural Networks. The “location”, “condominium fee”, and “property area” were the most employed features in each of the main categories: locational, financial, and physical, respectively. The main data sources for the ML models were real estate websites, which can present significant bias. A clear relationship was not observed between model quality metrics and the amount of data, nor the algorithm employed. The performance of the algorithms varied according to the database characteristics. The knowledge gaps identified are related to the impact of the time on property sale value, as well as the implementation of a correction factor that follows the sector’s temporality and improves future predictions.

Keywords Real Estate market · Machine learning · Property valuation · Property pricing · Literature review

Abbreviations

ADB	Adaboost
AI	Artificial Intelligence
ANN	Artificial Neural Networks

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COVID-19	Corona Virus Disease 2019
CTB	Catboost
DT	Decision Tree Regressor
EN	Elastic Net
ERT	Ensembles of Regression Trees
GBM	Gradient Boosting Machine
GP	Genetic Programming
HPM	Hedonic Price Modeling
KNN	K-Nearest Neighbors
LL	Linear Lasso
LR	Linear Regression
LRD	Linear Ridge
LSS	Lasso
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ML	Machine Learning
MLP	Multilayer Perceptrons
MLR	Multi Linear Regression
MSE	Mean Square Error
R^2	Coefficient of Determination
RDG	Ridge
RF	Random Forest
RMSE	Root Mean Square Error
SVM	Support Vector Machine
SVR	Support Vector Regression
XGB	XGBoost

1 Introduction

The real estate market has several stakeholders: for the majority of the population, property is an integral part of their daily lives; for investors, real estate represents a profitable asset; finally, the government generates income from property transactions (Kabaivanov & Markovska, 2021). Furthermore, the entire civil construction chain involves indirect stakeholders, including material manufacturers, construction companies, real estate agencies, property management companies, and building maintenance companies (Botelho, 2007, Temur et al. 2019).

According to Baldominos et al. (2018a, b), between 2016 and 2017, 1.1 trillion dollars were transacted by real estate management companies worldwide, demonstrating the economic significance of this sector. During the COVID-19 pandemic, central banks reduced these rates to support economies, which, in some regions of the world, contributed to boosting the real estate market (Neufeld, 2023). By the end of 2021, the global real estate sector was on the rise, with property prices growing at a faster rate than seen in 50 years (Romei & Smith, 2022). However, in less than a year, the situation changed dramatically due to the impacts of the war in Ukraine,

leading to a significant global slowdown and rising inflation worldwide (Irtysheva et al., 2022; Romei & Smith, 2022; Gabrielli et al., 2023).

In the real estate market, unlike traditional manufacturing industries, there is no standard method to price the products (Corrêa et al., 2018). The Law of Supply and Demand allows speculation on the value of traditional goods based on their market availability and manufacturing characteristics. However, for the real estate market, this analysis is more complex, with much more variables to be considered, such as the property's location, finishing standards, sunlight exposure, number of parking spaces, among other qualitative and quantitative factors. The desirability of these parameters is subjective, varying according to the buyer's preferences. To complicate things even further, the real estate sector is directly influenced by public policies, taxation, inflation, foreign scenario, and regulations at the municipal, state, and federal levels (Neto, 2015).

The heterogeneity of these variables strongly influences the property prices, which is a concern for all stakeholders (Rico-Juan & de La Paz, 2021; Ebekozen et al., 2019). Therefore, it is crucial to develop effective pricing methods and pinpoint the primary factors influencing these prices. These tools are essential for facilitating negotiations and establishing fair and competitive property prices (Zulkifley et al., 2020).

There are several studies that compare real estate pricing methods, identify and evaluate estimation tools, determine local (or submarket) indices, and develop new price estimation models (Krause, 2019). Some real estate pricing techniques are: the income approach; the cost approach; the comparative approach, and the hedonic price approach (Almaslukh, 2020). According to Almaslukh (2020), the income approach establishes property prices based on the expected future cash flow from the investment; the cost approach considers the property's value and depreciation over time; while the comparative approach suggests that similar properties should have similar prices. Hedonic model analysis identifies property characteristics and establishes a relationship between these characteristics and the final property price (Rico-Juan & de La Paz, 2021). Thus, hedonic regression models can be used to estimate the price of an asset with known characteristics (Baldominos et al., 2018a, b); being adaptable and able to capture factors that are difficult to quantify through other approaches (Mayer et al., 2019). In this context, many studies have assessed machine learning (ML) techniques for the prediction of property prices.

ML is a branch of artificial intelligence (AI) that involves computational algorithms that use prior knowledge (data) to generate predictions autonomously (without being explicitly programmed for it) (Mohri et al., 2018). This information may come in the form of human-labeled training sets or other types of data gathered through interactions with the environment, such as sensors (Bishop, 2006; Hastie et al., 2009; Mahesh, 2020). In all circumstances, the quality, representativeness, and volume of data are crucial for the success of ML models in making predictions (SAS, 2022).

In the past few years, these techniques have been vastly applied in the engineering field, to predict the compressive strength of concrete (Paixão et al., 2022); to detect damage in structures (Nunes et al., 2020); and optimize the design of geopolymer mixtures as well as the elastic modulus of geopolymeric composites (Ji et al.,

2024; Golafshani et al., 2024), which serve as alternative binders to Portland cement (Carvalho et al., 2023), among others. In the real estate market, these techniques can even help identify the variables that have the most impact on property pricing (Kabaivanov & Markovska, 2021).

The literature presents a few studies that evaluate ML methods for predicting real estate prices. For instance, Kuvalekar et al. (2020) compared the Support Vector Machine (SVM), Random Forest (RF), Linear Regression (LR), Multiple Linear Regression (MLR), Decision Tree Regressor (DT), and K-Nearest Neighbors (KNN) algorithms to properties in the city of Mumbai, India; with DT achieving the highest accuracy at 89%. Ho et al. (2020) compared SVM, RF, and Gradient Boosting Machine (GBM) algorithms to a database of over 40,000 real estate transactions in Hong Kong, China, and observed superior performances from RF and GBM.

In order to identify properties priced below the market, Baldominos et al. (2018a, b) employed Regression Trees (RT), KNN, SVM, and Multilayer Perceptrons (MLP) algorithms, with the RT presenting the best results. In turn, Castelli et al. (2020) employed Lasso, Ridge, Elastic Net, and Artificial Neural Networks (ANN) to identify the number of days that a given property remained online and, thus, assess the likelihood of such a listing being false. The authors observed an unethical practice among certain real estate agencies regarding fake advertisements, in which the property was not genuinely available or sometimes did not even exist. These deceptive ads are often employed solely to lure in customers to the agency's website.

In this sense, the present work aims to investigate the state of art in the application of ML algorithms in predicting house prices. The research also identified knowledge gaps, the most frequently adopted techniques, and the main characteristics inherent to them, in order to serve a basis for future researchers and practitioners.

So far, 4 literature review articles have been found regarding the use of ML in the real estate market for property pricing. The study conducted by Zulkifley et al. (2020) collected and evaluated 14 articles, extracting information related to the most commonly adopted input attributes. The authors also correlated the Root Mean Square Error (RMSE) values reported in some of the reviewed articles to the use of certain attributes. The review by Ja'afar et al. (2021) evaluated 24 articles, focusing on the application of machine learning techniques in real estate prediction and evaluation. It also aimed to identify the best model for predicting property prices based on factors such as location, land size, and the number of rooms. Al-Qawasmi (2022) analyzed 20 articles, providing a description of the adopted ML techniques. The author presented the key findings of the analyzed articles, grouping them by algorithms. Finally, Root et al. (2023) analyzed all the articles from the past two decades included in their systematic review, offering insights into how the use of ML has evolved over time in real estate market evaluation. This paper did not specify the number of articles evaluated, and the extracted information from the articles was also grouped by algorithm. All reviews concluded that ML models have the potential to contribute to property valuation and increase price accuracy. While there is not a single flawless ML technique, ongoing research can continue to fine-tune the methods employed and improve their quality.

In turn, the present study analyzed 63 articles, searching for research gaps. The only common aspects with the previous reviews are the identification of the adopted algorithms and the classification of input attributes. In this sense, the present review innovates

by identifying and correlating various pieces of information collected from the reviewed articles, analyzing parameters, models, and findings. The analysis also explored trends in model behavior based on the papers' scenarios, data volume, and type, as well as the results of model quality metrics (especially R^2). This examination revealed knowledge gaps that remained unaddressed in any of the previous review articles.

2 Methods

The current work has a descriptive nature, with data sources derived from the literature. Data collection procedures were carried out through systematic research, which steps are shown in Fig. 1.

The data collection was conducted through the Scopus database, as it is the largest global database of peer-reviewed literature abstracts and citations (Scopus, 2022). The search string adopted was: (*"Machine Learning" OR "data mining"*) AND (*"Real Estate Market" OR "Property Valuation" OR "house price" OR "building value"*). The last review for mapping new studies took place on August 31, 2023. Search filters included limiting the publication date from 2017 to 2023 to evaluate more recent studies; English language and Open Access. At this stage, both journal articles and conference papers published in journals were accepted. This initial search resulted in 125 articles.

Quality criteria (e.g., citations, authorship) were not applied to articles. The open access selection is a limitation of the present work, but it was a way to democratize access among researchers and practitioners.

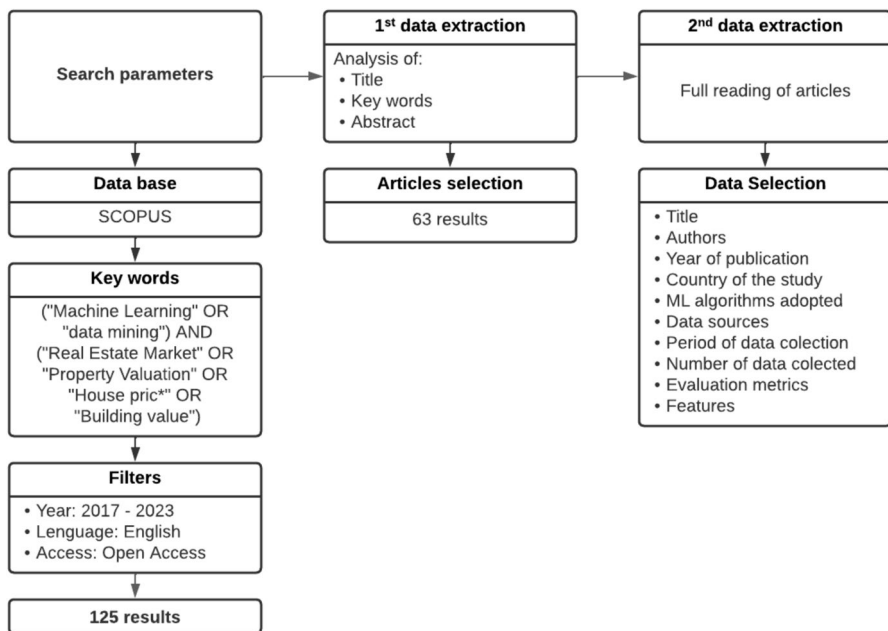


Fig. 1 Overview of the methodology

The next step involved the evaluation of the articles' titles, abstracts, and keywords, with the aim of excluding those not related to the application of ML algorithms in residential real estate pricing. Out of the 125 surveyed documents, 63 remained and were read in their entirety.

3 Results and Discussions

3.1 Summary of Results

To organize the gathered information, Table 1 includes the key data that will be presented individually in the subsequent sections, along with detailed discussions.

3.1.1 Countries with the Highest Research Activity in the Field

Figure 2 illustrates the main countries conducting research related to the application of ML in the real estate market (from Table 1). The studies encompass 21 countries across 4 continents, being 45% in Asia, 25% in Europe, 25% in the Americas, and 5% in Oceania. It is worth noting that the United States stands out, representing 22% of the studies assessed in this research.

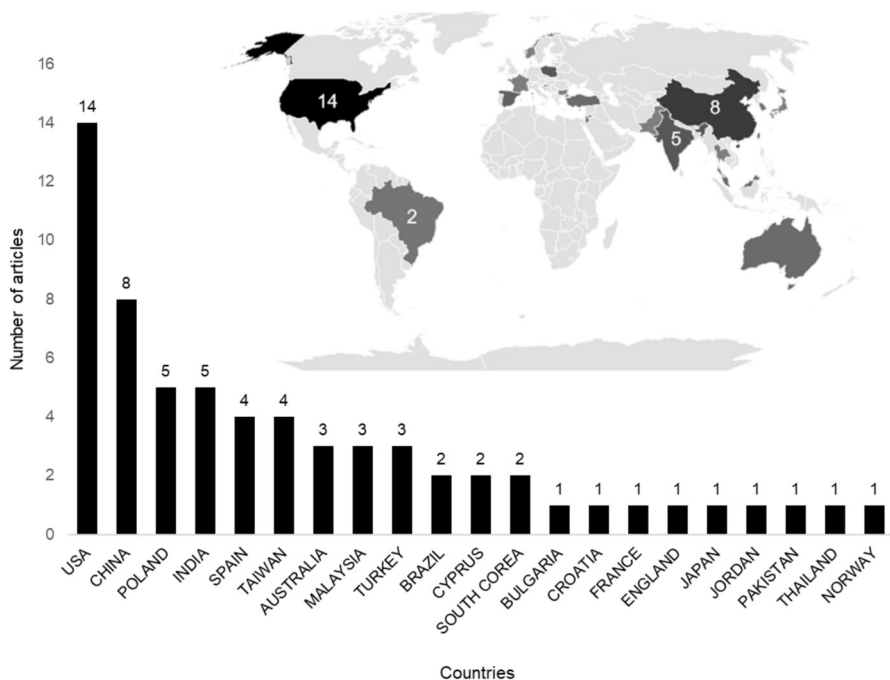


Fig. 2 Countries that published the analyzed articles. Source: Author

Table 1 Compilation of data collected in the review. Note: “N/I” means “not informed”

Citation	Year of Publication	Country	Algorithms adopted	Data Source	Period of data collection	Volume of data	Number of features
(Manjula et al., 2017)	2017	USA	Linear Regression Multivariate Regression Polynomial Regression Regression Tree Gradient Boosting Machine	N/I	N/I	21,000	15
(Gu & Xu, 2017)	2017	China	Linear Regression Artificial Neural Networks Pruned Model Tree	Real Estate Sales Website—Sina	N/I	253	14
(Trawiński et al., 2017)	2017	Poland	Linear Regression Artificial Neural Networks Pruned Model Tree	Public Directory of Real Estate Transactions	1998 to 2013	12,439	4
(Nejad et al., 2017)	2017	Australia	Random Forest Gradient Boosting Machine Boosted Tree	Directory of Real Estate Transactions—PriceFinder	May to November, 2015	1,967	4
(Oladunni et al., 2017)	2017	USA	Random Forest	Database Repository—Kaggle	2006 to 2017	15,827	-
(Trawiński et al., 2017a, b)	2017	Poland	Multilayer Perceptron Pruned Model Tree	Public Directory of Real Estate Transactions	1998 to 2013	12,439	4
(Baldominos et al., 2018a, b)	2018	Spain	Support Vector Regression K Nearest Neighbours Ensembles of Regression Trees	Real Estate Sales Website—Idealista	2nd semester of 2017	2,266	23
(Aderibigbe & Chi, 2018)	2018	USA	Artificial Neural Networks WEKA—R	Non-specified Real Estate Sales Website	1985 to 2017	-	-
(Phan, 2018)	2018	Australia	Linear Regression Regression Tree Artificial Neural Networks Support Vector Regression	Database Repository—Kaggle	2016 to 2018	34,857	11

Table 1 (continued)

Citation	Year of Publication	Country	Algorithms adopted	Data Source	Period of data collection	Volume of data	Number of features
(Varma et al., 2018)	2018	India	Linear Regression Forest Regression Boosted Regression Artificial Neural Networks	N/I	N/I	N/I	
(Zhao et al., 2018)	2018	USA	DCNN XGBoost Random Forest	Database Repository—Kaggle	N/I	20,000	5
(Ghosalkar & Dhage, 2018)	2018	India	Linear Regression	Non-specified Real Estate Sales Website	2005 to 2016	N/I	10
(Afonso et al., 2019)	2019	Brazil	Random Forest Kiss Model Ensemble Models	Non-specified Real Estate Sales Website	2015 to 2018	12,223,582	24
(García-Magarino et al., 2019)	2019	Spain	Linear Regression K Nearest Neighbours Support Vector Regression Artificial Neural Networks	Real Estate Sales Website—Idealista	2017	89	13
(Dimopoulos & Bakas, 2019a)	2019	Cyprus	Nonlinear Regression Linear Regression Random Forest	Public Directory of Real Estate Transactions de Cyprus	2008 to 2014	3,561	9
(Kumar et al., 2019)	2019	USA	Gradient Boosting Machine Decision Trees Artificial Neural Networks PCA	Public Directory—TerraFly,	N/I	4,500	9

Table 1 (continued)

Citation	Year of Publication	Country	Algorithms adopted	Data Source	Period of data collection	Volume of data	Number of features
(Masrom et al., 2019)	2019	Malaysia	Genetic Programming Random Forest Regressor Decision Tree Regressor Ridge Lasso Linear Regression	N/I	N/I	4,066	N/I
(Mohd et al., 2019)	2019	Malaysia	Random Forest Regressor Decision Tree Regressor Ridge Lasso	N/I	2016	4,066	19
(Lydia et al., 2019)	2019	USA	Linear Regression Linear Regression	Real Estate Deliberation Company in Boston	N/I	N/I	14
(Tang et al., 2018)	2019	China	K Nearest Neighbours Gradient Boosting Machine	Real Estate Sales Website -Fang.com	1 st semester of 2018	N/I	21
(Dimopoulos & Bakas, 2019b)	2019	Cyprus	Univariate Multivariate Linear Higher Order Regression	Public Directory of Real Estate Transactions de Cyprus	-	3,561	12
(Talaga et al., 2019)	2019	Poland	Artificial Neural Networks Random Forest Linear Regression Decision Tree Artificial Neural Networks Decision Trees XGBoost	Public Directory of Real Estate Transactions	2007–2015	21,000	25

Table 1 (continued)

Citation	Year of Publication	Country	Algorithms adopted	Data Source	Period of data collection	Volume of data	Number of features
(Ye et al., 2019)	2019	China	Linear Regression	Real Estate Sales Website—Homelink	2016	1,395	N/I
(Niu & Niu, 2019)	2019	China	Gradient Boosting Machine Random Forest	Real Estate Company Data	June to september of 2017	44,113	4
(Almaslukh, 2020)	2020	Taiwan	Artificial Neural Networks Gradient Boosting Machine XGBoost Support Vector Regression	Database Repository—UCI	N/I	414	5
(Kuvalekar et al., 2020)	2020	India	Support Vector Regression Random Forest Linear Regression Multiple Linear Regression Decision Tree Regressor	Non-specified Real Estate Sales Website	N/I	N/I	9
(Castelli et al., 2020)	2020	Bulgaria	K Nearest Neighbours Random Forest Elastic Net Lasso Ridge Artificial Neural Networks K Nearest Neighbours	Real Estate Sales Website in Sofia—Homehead	2015 to 2018	550,000	20
(Ho et al., 2020)	2020	China	Support Vector Regression Support Vector Regression Random Forest Gradient Boosting Machine	Public Directory of Real Estate Transactions from Hong Kong	1996 to 2014	39,554	6

Table 1 (continued)

Citation	Year of Publication	Country	Algorithms adopted	Data Source	Period of data collection	Volume of data	Number of features
(Al-Sit & Al-Hamad, 2020)	2020	Jordan	Linear Regression Gradient Boosting Machine Support Vector Regression Random Forest	Non-specified Real Estate Sales Website	03/05/2020 a 04/28/2020	3,697	25
(Pai & Wang, 2020)	2020	Taiwan	LSSVR CART GRNN BPNN	Public Directory of Real Estate Transactions from Taichung	2016 to 2019	32,215	23
(Ahtesham et al., 2020)	2020	Pakistan	Gradient Boosting Machine	Public Directory of Real Estate Transactions from Pakistan	N/I	168,447	17
(Tejaswi & Satyanarayana, 2020)	2020	India	Decision Tree Regression Linear Regression Ada Boost Regression Gradient Boosting Machine Random Forest Regression	Database Repository—Kaggle	N/I	N/I	70
(Levantessi & Piscopo, 2020)	2020	England	Random Forest	Public Directory of Real Estate Transactions from Londres	1996 to 2018	N/I	7
(Jain et al., 2020)	2020	USA	N/I	N/I	N/I	N/I	N/I
(Kang et al., 2020)	2020	USA	Gradient Boosting Machine Multi Linear Regression	Non-specified Real Estate Sales Website	2014 to 2019	21,928	8
(Jarosz et al., 2020)	2020	Poland	Gradient Boosting Machine Multi Linear Regression Random Forest Bagging Regressor	Cadastral system in Poland	1998 to 2015	83,523	4

Table 1 (continued)

Citation	Year of Publication	Country	Algorithms adopted	Data Source	Period of data collection	Volume of data	Number of features
(Mrsic et al., 2020)	2020	Croatia	Linear Regression Decision Tree Gradient Boosting Machine AdaBoost	Non-specified Real Estate Sales Website	N/I	7,416	41
(Chiramel et al., 2020)	2020	USA	XGBoost Linear Regression Ridge Regularization	Public Directory of Real Estate Transactions from Iowa	N/I	2,920	81
(Guo et al., 2020)	2020	China	Random Forest	Public Directory of Real Estate Transactions of China	2011 to 2017	N/I	N/I
(Chanasit et al., 2021)	2021	Thailand	Artificial Neural Network	Database Repository—Kaggle	2016 to 2017	90,275	58
(Abdul-Rahman et al., 2021)	2021	Malaysia	XGBoost Light Gradient Boosting Multi Linear Regression	Database Repository—Kaggle	2019	53,883	12
(Kabaivanov & Markovska, 2021)	2021	USA	Ridge Regression Artificial Neural Networks	NAREIT e BGREIT—real estate market organizations	1972 to 2020	582	N/I

Table 1 (continued)

Citation	Year of Publication	Country	Algorithms adopted	Data Source	Period of data collection	Volume of data	Number of features
(Rico-Juan & de La Paz, 2021)	2021	Spain	K Nearest Neighbours Decision Tree Random Forest AdaBoost XGBboost CatBoost Artificial Neural Networks Linear regression Linear Ridge Linear Lasso	Tabimed (important real estate appraisal company that has been dissolved)	2004 to 2012	392,412	53
(Tchuente & Nyawa, 2021)	2021	France	Artificial Neural Networks Random Forest Gradient Boosting Machine K Nearest Neighbours Linear Regression Support Vector Regression	Public Directory of the French Government	2015 to 2019	N/I	9
(Li et al., 2021)	2021	China	HPM XGBboost	Chinese Real Estate Information Platform	N/I	12,137	35
(Peng, 2021)	2021	Japan	Decision Tree Random Forest	Ministry of Land, Infrastructure, Transport, and Tourism of Japan	2016 to 2019	16,276	25

Table 1 (continued)

Citation	Year of Publication	Country	Algorithms adopted	Data Source	Period of data collection	Volume of data	Number of features
(Marzagão et al., 2021)	2021	Brazil	Linear Regression Random Forest Artificial Neural Networks Boosted Trees	Real Estate Sales Website in Brasilia	N/I	15,552	21
(Rizun & Baj-Rogowska, 2021)	2021	Poland	Support Vector Regression Regularized Regression Support Vector Regression K Nearest Neighbours Naive Bayes CART	Website Trading Economics	2011 to 2019	N/I	X
(Lee, 2022)	2022	South Korea	Random Forest Artificial Neural Networks	Non-specified Real Estate Sales Website	N/I	22,213	6
(Tekin and Sari, 2022)	2022	Turkey	Linear Regression Polynomial Regression Decision Trees Random Forest XGBoost	Non-specified Real Estate Sales Website	2020	22,176	11

Table 1 (continued)

Citation	Year of Publication	Country	Algorithms adopted	Data Source	Period of data collection	Volume of data	Number of features
(Gao et al., 2022)	2022	Australia	Lasso Regression Ridge Regression Elastic Net Support Vector Regression Decision Trees Random Forest XGBoost	Non-specified Real Estate Sales Website	2018	22,979	20
(Mora-Garcia et al., 2022)	2022	Spain	Artificial Neural Networks Linear Regression	Non-specified Real Estate Sales Website	2019 to 2021	39,943	27
(Hjort et al., 2022)	2022	Norway	Linear Regression	N/I	2013 to 2015	126,719	26
(Zaki et al., 2022)	2022	USA	XGBoost	Database Repository—Kaggle	N/I	506	14
(Yağmur et al., 2023)	2022	Turkey	Artificial Neural Networks	Non-specified Real Estate Sales Website	august/22	900	6
(Singh et al., 2022)	2022	India	Random Forest Artificial Neural Networks Gradient Boosting Machine Support Vector Regression Multi Linear Regression	Database Repository—Kaggle	N/I	7,109	19
(Teoh & Yau, 2022)	2022	USA	Linear Regression Random Forest	Database Repository—Kaggle	N/I	2,930	15

Table 1 (continued)

Citation	Year of Publication	Country	Algorithms adopted	Data Source	Period of data collection	Volume of data	Number of features
(Chou et al., 2022)	2022	Taiwan	Artificial Neural Networks Support Vector Regression	Public Directory of Real Estate Transactions	2013 to 2017	13,220	27
(Kim et al., 2022)	2022	South Korea	Linear Regression Artificial Neural Networks	Public Directory of Real Estate Transactions	2015 to 2019	287,034	16
(Choy & Ho, 2023)	2023	China	Random Forest Extra Trees K Nearest Neighbours	Public Directory of Real Estate Transactions	1997 to 2021	24,317	10
(Çilgım & Gökçen, 2023)	2023	Turkey	Random Forest Linear Regression Lasso Regression Ridge Regression XGBoost	Non-specified Real Estate Sales Website	07/06/2021	18,420	86
(Lahmiri et al., 2023)	2023	Taiwan	Artificial Neural Networks Ensemble Regression Trees Support Vector Regression	Public Directory of Real Estate Transactions	2012 to 2013	414	5
(Sibindi et al., 2023)	2023	USA	Gaussian Process Adaboost GBM LGBM XGBoost LGBM-XGBoost	Public Directory of Real Estate Transactions	1990	20,640	9

The diversity of countries in which these studies have been conducted emphasizes the significant need for robust and reliable models for property pricing, and how widely the topic of ML is spread throughout the world. Both phenomena probably derive from the real estate sector's great capacity to drive and boost the economy, as it is connected to several other branches of activities and services worldwide.

Figure 3 shows that the number of publications in the field has been increasing each year, reinforcing that this topic is a big concern for those involved in the real estate sector. There is a noticeable decrease in the year 2021, followed by growth again in 2022. Possible justifications for this trend could include the COVID-19 pandemic and the fact that the research was limited to articles in English and freely accessible.

The dilemma of property pricing is highly complex and involves multiple quantitative and qualitative variables, as well as very extensive databases. Therefore, the use of AI for addressing these issues is quite promising, as shown by the large number of publications in the area (Zulkifley et al., 2020; Ja'afar et al. 2021; Al-Qawasmi, 2022; Root et al., 2023).

3.1.2 Data Sources and Dataset Characteristics

The diversity of data sources is a notable characteristic observed in Table 1. Out of the 63 analyzed studies, 22 used real estate sales websites (from which 13 did not specify which website); 19 used directories from public agencies that store data related to real estate transactions; 6 collected data through online platforms of companies and organizations that store real estate sector data; 11 adopted databases from public domain repositories used for data science studies and competitions, and 5 studies did not specify the source of data used in their research.

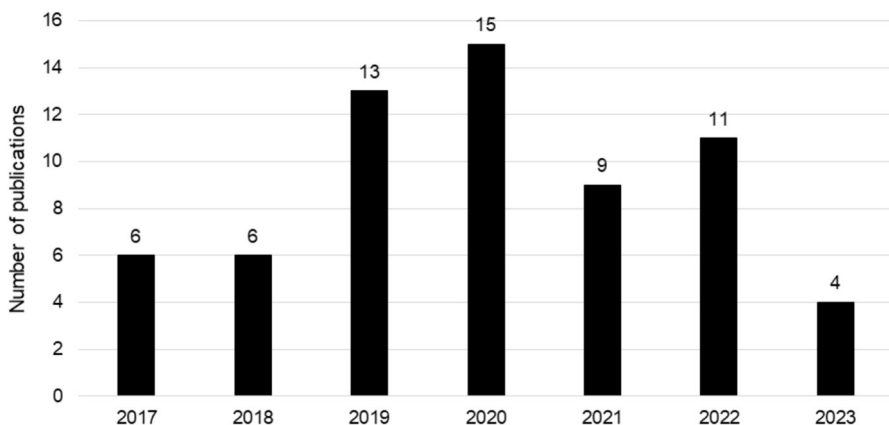


Fig. 3 Number of publications per year, up to August 31 st, 2023. Source: Author

In these studies, 53% created their own database using properties that had actually been sold, while the remaining 47% worked with properties listed for sale, which were collected from online platforms. This difference is relevant, because the properties' listing prices may significantly differ from their effective transaction values, as negotiation is a common practice in this type of commerce. This factor can affect the accuracy of predictive models that rely solely on listed data.

Another crucial issue from online platforms are the fake listings, as mentioned by Castelli (2020). The potential existence of fraudulent listings within the databases impacts the predictive models, which will use data from non-existent properties to identify patterns for future predictions.

But more serious than that are the works that do not even mention the data source, or only say that the data was gathered from an undefined "property sales website". Lack of clarity in data makes it difficult to assess the quality and representativeness of training datasets (Caton & Haas, 2023). Reproducibility in ML research is essential for validating results and for policymakers to build trust in the scientific community. The lack of transparency in data processing and algorithms can make it difficult to interpret and understand how models make decisions, compromising the reliability of the research (Mehrabi et al., 2021; Caton & Haas, 2023).

Figure 4 displays the distribution of the number of articles by number of instances collected, demonstrating that the highest frequency is in the range of 20,000 to 50,000. Out of the 14 works detected within this data volume range, 7 sourced their data from advertisements, 6 from sold properties, and 1 did not specify the data source.

A higher volume of information is commonly associated with improved quality in ML algorithms, because large datasets allow the algorithms to analyze and learn from a greater diversity of features to identify patterns for future predictions (Paixão et al., 2022; Luca et al., 2022). In this regard, only 6 (9.5%) of the 63 analyzed articles evaluated more than 100,000 learning instances, while 7 others (11%) worked with fewer than 1,000. Although

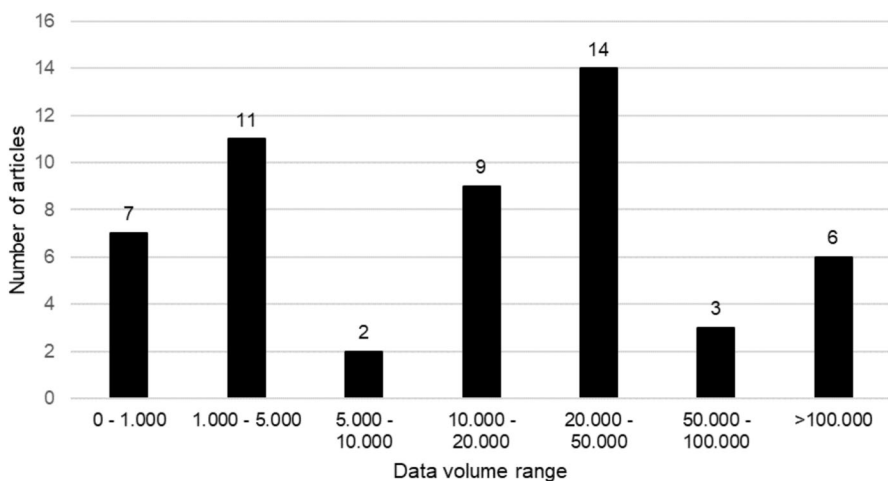


Fig. 4 Number of studies per range of data collected. Source: Author

more instances are preferable, the size of the dataset is not the only determining factor in algorithm performance. The prediction quality is also influenced by the representativeness of the samples (if there are enough instances from all categories), the presence of outliers and data gaps; the complexity of the problem, and the algorithm's ability to learn from the available data (Sutton & Barto, 2018). Therefore, even studies that used fewer than 1,000 instances can still yield good accuracy, if their dataset has enough quality.

From the analyzed studies, 7 collected data at a national level; 5 used data collected at the state level, encompassing multiple cities; and the remaining 51 employed data collected from a single city. This distribution is relevant because an imbalanced multiarea database (i.e., where areas are not evenly represented) may influence the model's performance. Another consideration is that in highly variable datasets, some attributes may have a more significant impact on the target variable (in this case, the price), whereas in a lower variability setting, these attributes may not have such an influence (Bishop, 2006).

As seen in Table 1, the data collection period of many works was relatively long, being subjected to external influences related to politics, inflation, and taxes. However, out of the 63 reviewed articles, only the studies of Ho et al. (2020) and Dimopoulos and Bakas (2019a, 2019b) applied/tested a correction factor to update the house prices. Yet, none of them compared the results of the updated prices with a simulation without applying the index. It might be interesting to evaluate and compare both situations to understand how the algorithms perform with inflation and correction indexes.

3.1.3 Most Adopted Features

Summing up all articles, 26 features were identified when assessing residential property prices. Figure 5 illustrates the features with three or more mentions. Using this information, these features fell into three categories: locational, financial, and physical.

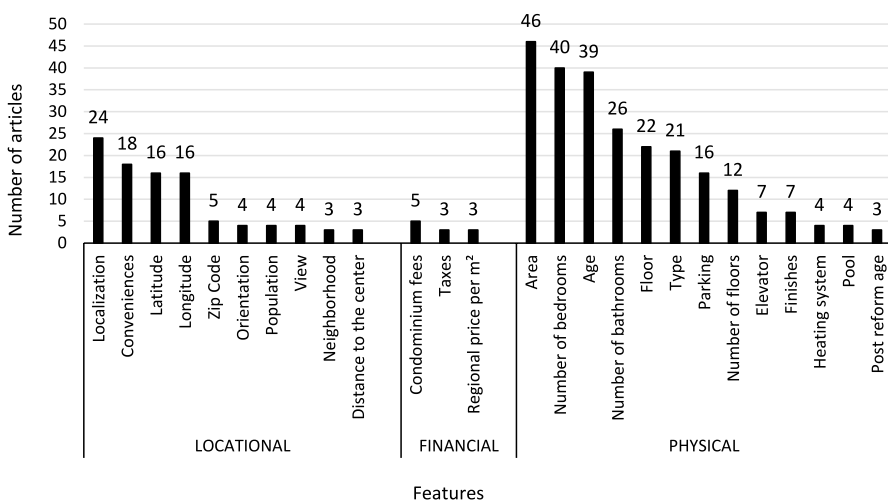


Fig. 5 Most adopted input features in property pricing models. Source: Author

The first category, "Locational," encompasses features related to the property's address and neighborhood. Location is considered the most significant feature in determining property prices by some authors (Gao et al., 2019) (Rahadi et al. 2016) (Alfiyatin et al. 2017). The relevance of this feature is linked to the fact that certain areas within a city have better infrastructure, traffic, and security, consequently leading to higher values (Grybauskas et al. 2021). To evaluate the Locational features, some studies provided the complete property address, others summarized the information using latitude and longitude coordinates, others used only the ZIP Code, and others evaluated it as the distance to the city center. This discrepancy of approaches is worth noticing, and no comparison studies were found to investigate which ones are the most effective.

Figure 5 also shows that the view from the property (whether it is of natural or modified landscape) is not frequently considered in studies on this topic, as well as the orientation in relation to the sun and the population of the area where the property is situated.

The second category, "Financial", groups the features related to financial expenses associated with the acquisition, use, and maintenance of the property. These items encompass taxes, condominium fees, and the average price per square meter found in the region where the property is located. They were not mentioned as many times as the Locational and Physical features.

Finally, the category identified as "Physical" represents attributes provided by builders or developers to attract potential buyers to acquire the property, such as floor area, number of bedrooms and number of parking spots. In their study, Owusu-Manu et al. (2019) observed that physical parameters are just as influential do the price as the locational ones. Zhou (2020) indicates that the 3 main attributes influencing residential property prices are the property's size, the total number of bathroom units, and their overall quality (materials and finishes). Following a similar approach, Owusu-Manu et al. (2019) emphasized that adding a floor, a bedroom, and a bathroom increases the property's value by 13%, 16%, and 2%, respectively.

3.1.4 Most Adopted Machine Learning Algorithms

Just as it is crucial to assess the primary factors affecting property prices, it is equally important to pinpoint the most suitable tools capable of utilizing such input data to accurately predict those prices. The analysis of the 63 reviewed articles identified the most frequently used ML algorithms, as summarized in Fig. 6.

Adhering to the criterion of a minimum frequency of 3 occurrences, 11 models were the most adopted: RF (Random Forest), Extreme Gradient Boosting Machine (XGBoost) + Gradient Boosting Machine (GBM), LR (Linear Regression), ANN (Artificial Neural Networks), SVR (Support Vector Regression), DT (Decision Tree Regressor), KNN (K Nearest Neighbors), RIDGE, AdaBoost (Adaptive Boosting),

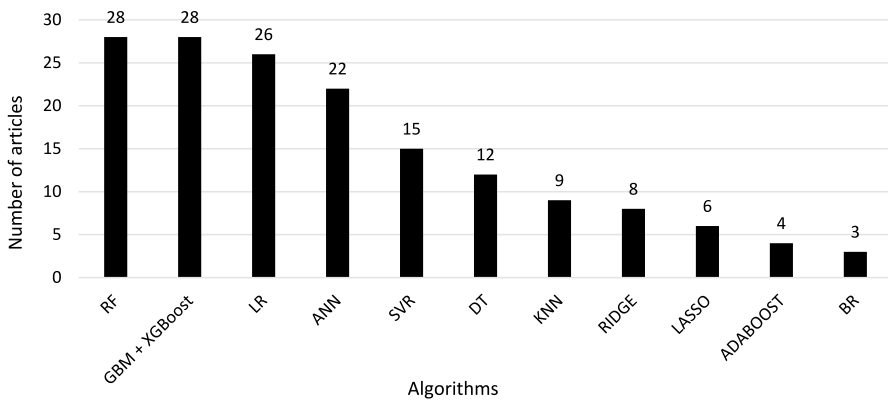


Fig. 6 Most adopted machine learning algorithms in property pricing

LASSO (Least Absolute Shrinkage and Selection Operator), and BR (Boosted Regression). These algorithms can be grouped by the method used for data training and testing:

- Decision tree-based models: RF, XGBoost, GBM, DT, AdaBoost, and BR.
- Linear regression: LR, RIDGE, and LASSO.
- Others: ANN, SVR, and KNN.

The Gradient Boosting Machine (GBM) presents high performance, consistency, and interpretability (Zhou, 2020). It produces a predictive model in the form of an ensemble of weak predictive models, typically decision trees (Ho et al. 2020). After identifying the weak learnings in each regression tree, boosting assigns equal weights to each observation, focusing on the errors between the values predicted by the weak learning and the actual values (Ho et al. 2020). XGBoost and GBM are both boosting-based machine learning algorithms that use decision trees as base models. XGBoost is an evolution of GBM as it incorporates regularization terms to prevent overfitting, thereby enhancing the model's generalization capacity (Zhou, 2020). XGBoost is highly optimized for parallel processing, allowing faster training on large datasets and efficiently handling missing values in training data, making it more robust for real-world scenarios (Chen & Guestrin, 2016). Therefore, in this research, the GBM and XGBoost were grouped into the same class of algorithms.

The random forest (RF) is a supervised algorithm that employs ensemble learning methods for classification and regression. RF builds n regression trees and combines them into a single model to make more accurate predictions than a single tree (Silva, 2019). Each tree can cast one vote for the value that best represents a given input, and the final model output will be the most representative value voted on by the n trees in the ensemble (Afonso et al. 2019) (Silva, 2019).

According to Stevenson (1981), linear regression (LR) is based on the analysis of data in order to determine the relationship between dependent and independent variables, with the result of this analysis being an equation that describes the relationship between the variables in mathematical terms. In a LR model, the predicted property price will be a linear combination of its features. However, this model is susceptible to random errors when input parameters exhibit a high degree of collinearity, and cannot capture non-linear behavior (García-Magariño et al. 2019). Multiple regression models can determine which variables or property characteristics are more relevant in explaining the dependent variable (Zulkifley et al. 2020).

The Artificial Neural Network (ANN) model is used to capture nonlinear relationships between input and output data, showing promise in estimating residential property prices (Zulkifley et al. 2020). Inspired by the human brain, this model consists of layers of interconnected nodes (neurons). Each node processes inputs, applies weights, and passes the results to the next layer. Learning occurs by adjusting the weights through techniques like backpropagation to minimize the error between predicted and actual outputs. However, results may vary significantly depending on the dataset used for training (Silva, 2019).

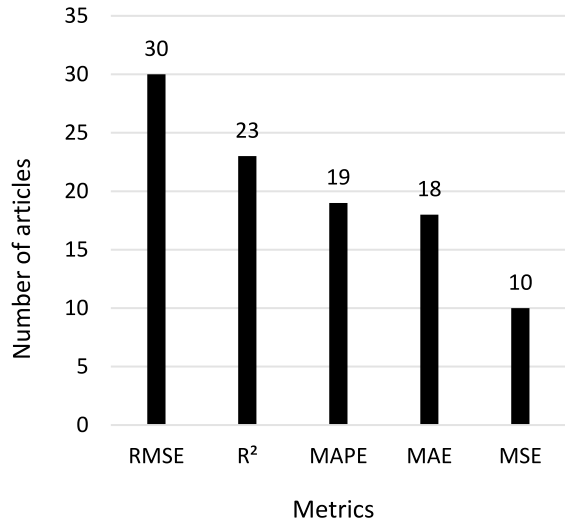
Support Vector Machines (SVM) are supervised learning techniques originally developed to handle pattern categorization problems and subsequently extended to modeling data. In this context, they are referred to as Support Vector Regression (SVR) (Smola & Scholkopf, 2003). The SVR learning method is based on the statistical learning theory of Vapnik, created in 1995, with the goal of providing ML strategies that maximize generalization (Acosta et al. 2016). According to these authors, SVR are used to find a predictor that closely approximates the sample data.

It is worth noting that these models incorporate both qualitative attributes (categorical variables), such as floor type, neighborhood name, and parking type, as well as quantitative attributes (numerical variables), such as the number of bedrooms, number of bathrooms, and floor area. However, no ML algorithm deals directly with categorical variables – they must be directly or indirectly converted to numbers for the algorithms to operate them – a process called “encoding”. Only a few authors, such as (Ho et al. 2020, Gu & Xu, 2017, García-Magariño et al. 2019, Rizun & Baj-Rogowska, 2021) and (Grybauskas et al. 2021), mentioned that they dealt with categorical variables using a binary treatment, while others did not mention the applied encoding technique.

From the collected data, it is evident that authors commonly adopt and compare more than one ML algorithm in each paper, since evident that the calculation strategy adopted by each algorithm can be more or less adaptable to the characteristics of each dataset. No work was found comparing the various existing encoding techniques for this type of problem, which can significantly affect the accuracy of the models.

3.1.5 Most Adopted Metrics for Assessing the Quality of ML Models

The adoption of a particular algorithm does not necessarily lead to a good prediction quality, which can be evaluated using different types of metrics. Figure 7 presents the metrics that have been most commonly adopted in the studies, considering

Fig. 7 Model quality assessment metrics

a minimum frequency of 3. The Root Mean Square Error (RMSE), the Coefficient of Determination (R^2), and the Mean Absolute Percentage Error (MAPE) were the most frequently employed metrics, followed by the Mean Absolute Error (MAE) and the Mean Square Error (MSE).

These metrics relate the predicted values, $\hat{y}_i$, to the actual (observed) values, y_i , and to the mean of the observations, $\bar{y}_i$, for the period i , where $i = 1, 2, \dots, n$, as follows:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (1)$$

$$R^2(y, \hat{y}) = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y}_i)^2} \quad (2)$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (3)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (4)$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (5)$$

R^2 expresses the percentage of the variation in the response variable that is explained by the variation in the explanatory variable (Martins 2018). As presented by Baldominos et al. (2018a, 2018b), R^2 provides a measure of how well future

samples are likely to be predicted. The R^2 value varies from negative infinity to 1. If the evaluated model results in a fit worse than the line representing the mean value of the data, R^2 will be negative (Paixão et al., 2022). Values close to 0 indicate low explanation of data variability, while values close to 1 indicate that the model fits the data well and can effectively predict the dependent variable(s) (James et al., 2017).

While the R^2 is a commonly used metric in the ML field, relying solely on it can pose challenges in drawing comprehensive conclusions about the model's performance. Therefore, it is advisable to utilize other metrics, such as MAE, which provides the average error in both positive and negative directions; and RMSE, which not only offers an average of errors but also penalizes larger errors more severely, thereby highlighting the impact of large errors on the model's predictions (Júnior, 2021). MAPE expresses the error as a percentage of the average response, which allows for an easier comparison among different algorithms regardless of the scale of the data. For this reason, it is always recommended that authors include at least MAPE besides R^2 . For MAE, RMSE and MAPE, lower values are desirable because they indicate lower differences between predicted and actual values.

3.1.6 Relationship Between the Quality of the Model and the Characteristics of the Dataset

Table 2 displays the distribution of R^2 values obtained for each algorithm in each of the 23 studies that adopted it to evaluate the price prediction.

From information in Table 2, Fig. 8 presents a scatterplot relating each algorithm to its respective R^2 value and the volume of data collected in the study. The graph shows that an increase in data volume does not necessarily result in a higher R^2 value, and thus, better model performance.

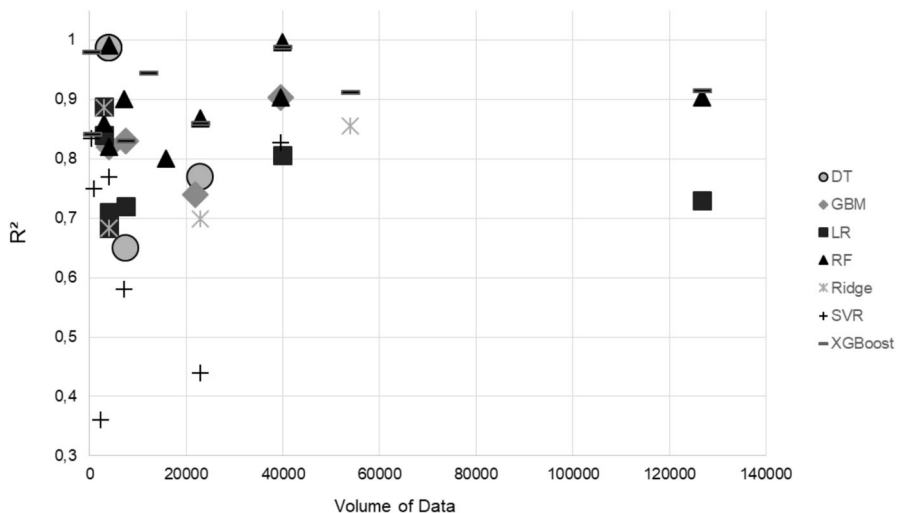


Fig. 8 Relationship between the volume of data and the R^2 of each study. Source: Author

Table 2 Coefficient of Determination (R^2) obtained in each study vs the adopted algorithm

Article Algorithm	(Oladunni et al., 2017)	(Baldomino et al., 2017)	(Masrom et al., 2019)	(Mohd et al., 2019)	(Almaslakh, 2020)	(Kuvalekar et al., 2020)	(Castelli et al., 2020)	(Al-Sit & Al-Hamadin, 2020)	(Mrsic et al., 2020)	(Chiramel et al., 2020)	(Rico-Juan & de La Paz, 2021)	(Ho et al., 2020)
ADB	-	-	-	-	-	-	-	-	0.820	-	0.667	-
ANN	-	0.307	-	-	-	-	0.200	0.730	-	-	0.886	-
CTB	-	-	-	-	-	-	-	-	-	-	0.935	-
DT	-	-	0.986	0.986	0.910	0.890	-	-	0.650	-	0.747	-
EN	-	-	-	-	-	-	0.198	-	-	-	-	-
ERT	-	0.125	-	-	-	-	-	-	-	-	-	-
GBM	-	-	-	-	0.985	-	-	0.820	0.830	-	-	0.904
GP	-	-	0.995	-	-	-	-	-	-	-	-	-
HPM	-	-	-	-	-	-	-	-	-	-	-	-
KNN	-	0.36	-	-	-	-	0.239	-	0.650	-	0.928	-
LSS	-	-	0.683	0.683	0.960	-	0.183	-	-	-	-	-
LL	-	-	-	-	-	-	-	-	-	-	0.423	-
LR	-	-	0.683	0.683	-	-	-	0.710	0.720	0.888	0.823	-
LRD	-	-	-	-	-	-	-	-	-	-	0.823	-
MLR	-	-	-	-	-	-	-	-	-	-	-	-
RF	0.800	-	0.991	0.991	0.970	-	0.363	0.820	-	-	0.986	0.903
RDG	-	-	0.683	0.683	-	-	0.194	-	-	0.886	-	-
SVR	-	-0.066	-	-	0.834	-	0.076	0.770	-	-	-	0.827
XGB	-	-	-	-	0.980	-	-	-	0.830	-	0.897	-

Table 2 (continued)

Article Algorithm	Li et al., 2021	(Kang et al., 2020)	(Choy & Ho, 2023)	(Gao et al., 2022)	(Mora-Garcia et al., 2022)	(Hjort et al., 2022)	(Zaki et al., 2022)	(Yagmur et al., 2023)	(Singh et al., 2022)	(Teoh & Yau, 2022)	(Abdul- Rahman et al., 2021)
ADB	-	-	-	-	-	-	-	-	-	-	-
ANN	-	-	-	0.849	-	-	-	0.827	0.600	-	-
CTB	-	-	-	-	-	-	-	-	-	-	-
DT	-	-	-	0.771	-	-	-	-	-	-	-
EN	-	-	-	0.419	-	-	-	-	-	-	-
ERT	-	-	-	-	-	-	-	-	-	-	-
GBM	-	0.740	-	0.867	0.993	-	-	-	0.650	-	-
GP	-	-	-	-	-	-	-	-	-	-	-
HPM	0.600	-	-	-	-	-	-	-	-	-	-
KNN	-	-	0.895	-	-	0.734	-	-	-	-	-
LSS	-	-	-	0.419	-	-	-	-	-	-	-
LL	-	-	-	-	-	-	-	-	-	-	-
LR	-	-	-	-	0.805	0.730	-	-	-	0.840	-
LRD	-	-	-	-	-	-	-	-	-	-	-
MLR	-	0.480	-	-	-	-	-	0.721	0.549	-	0.871
RF	-	-	0.919	0.868	0.997	0.904	-	-	0.901	0.860	-
RDG	-	-	-	0.699	-	-	-	-	-	-	0.855
SVR	-	-	-	0.439	-	-	-	0.749	0.580	-	-
XGB	0.944	-	-	0.859	0.988	0.914	0.841	-	-	-	0.912

ADB AdaBoost, ANN Artificial Neural Networks, CTB Catboost, DT Decision Tree, EN Elastic Net, ERT Ensembles of Regression Trees, GBM Gradient Boosting Machine, GP Genetic Programming, HPM Hedonic Price Modeling, KNN K Nearest Neighbours, LSS Lasso, LL Linear Lasso, LR Linear Regression, LRD Linear Ridge, MLR Multi Linear Regression, RF Random Forest, RDG Ridge, SVR Support Vector Regression, XGB XGBoost

The absence of a clear trend in the behavior of each algorithm aligns with the idea that there are several factors influencing the performance of ML models besides the algorithm. The quality of data is as crucial as its quantity. Many of the observations can be outliers (atypical values) that significantly influence prediction accuracy. Additionally, databases with many instances sharing similar characteristics yield poorer predictions compared to databases with variability in the input and output attributes, since the latter allow the algorithm to learn from a wider range of data. Finally, unbalanced databases, with some categories being more represented than others, can also impair model performance.

To enhance the quality of the data, several preprocessing steps can be applied, such as outlier removal, data scaling or normalizing, techniques to reduce or expand the observations used, techniques to fill in missing values, among others. Again, it was noticed that authors almost never clearly presented all the treatments adopted, making it difficult to understand the relationship between the resulting metrics and the dataset characteristics. Therefore, based on the collected information and the conducted analyses, it is not possible to indicate which algorithms demonstrate the best performance and which should be adopted in similar studies for predicting property sales value.

In a numerical analysis of the presented results, the highest R^2 found was 0.997 in the study conducted by Mora-Garcia et al. (2022), with the RF algorithm. This study examined 39,943 instances collected in the city of Alicante, Spain, from online real estate listings, and applied 27 attributes for price prediction. On the other hand, the study conducted by Almaslukh (2020), which used 414 instances (1.03% of the dataset size in Mora-Garcia et al. (2022)) and only 5 attributes, yielded an R^2 of 0.985 with the GBM algorithm. GBM was also employed in the study by Ho et al. (2020), who collected 39,554 instances, resulting in an R^2 of 0.904. These different R^2 results underscore the notions that the quality of the model is not directly related to the volume of data and that determining which ML algorithm performs best is not a simple task.

Comparison of the data in Table 2 with the data collection levels in Table 1 (municipal, state, or national level) does not reveal a clear relationship between the range of the data and the R^2 values. It would be expected that national-level studies would exhibit greater variability of characteristics and consequently lower R^2 values than city-level ones. However, this expectation was not met within the reviewed articles.

Tables 3 and 4 present, respectively, the values of MAE and RMSE obtained in the analyzed studies for each algorithm. Since both metrics measure in absolute values and considering differences in the magnitudes of property values in different cities, it is only possible to compare the MAE and RMSE values between different algorithms used in the same studies.

Table 3 shows that out of 5 studies that tested the RF algorithm alongside other models, 4 of them found that RF yielded the best MAE results. The same pattern emerged when examining the RMSE values in Table 4, which out of the 11 articles that compared RF to other models, 7 reported this algorithm having the lowest RMSE values. This observation is an indication that the RF algorithm has adapted well to the data used for model training and testing, which

Table 3 Ratio of the Mean Absolute Error (MAE) obtained in each study. Source: Author

Article Algorithm	(Almaslukh, 2020)	(Baldominos et al., 2018a, 2018b)	(Rico-Juan & de La Paz, 2021)	(Castelli et al., 2020)	(Tchuente & Nyawa, 2021)	(Ahleshham et al., 2020)	(Li et al., 2021)	(Marzagão et al., 2021)	(Dimopoulos & Bakas, Artificial intelligence for mass appraisals of residential properties in Nicosia: Math- ematical modelling and algorithmic implementation, 2019a)	(Nejad et al., 2017)
ADB	-	-	0.272	-	66,243	-	-	-	-	-
ANN	-	0.738	0.155	0.034	55,290	-	-	11.08	-	-
BT	-	-	-	-	-	-	-	9.49	-	-
CAT	-	-	0.12	-	-	-	-	-	-	-
DT	-	-	0.234	-	-	-	-	-	-	-
EN	-	-	-	0.034	-	-	-	-	-	-
ERT	-	-	0.3714	-	-	-	-	-	-	-
GBM	0.008	-	-	-	59,822	22,502	-	-	24,486	0.067
HPM	-	-	-	-	-	-	0.117	-	-	-
KNN	-	0.563	0.076	0.033	57,263	-	-	-	-	-
LASSO	-	-	-	0.034	-	-	-	-	-	-
LL	-	-	0.49	-	-	-	-	-	-	-
LR	-	-	0.2	-	64,901	-	-	13.49	22,978	-
LRIDGE	-	-	0.2	-	-	-	-	-	-	-
MLR	-	-	-	-	-	-	-	-	22,526	-
RF	-	-	0.03	-	56,246	-	-	8.14	20,817	0.061
RIDGE	-	-	-	0.034	-	-	-	-	-	-
SVR	0.035	0.440	-	0.039	68,226	-	-	11.36	-	-

Table 3 (continued)

Article Algorithm	(Almaslukh, 2020)	(Baldominos et al., 2018a, 2018b)	(Rico-Juan & de La Paz, 2021)	(Castelli et al., 2020)	(Tchuente & Nyawa, 2021)	(Ahtesham et al., 2020)	(Li et al., 2021)	(Marzagão et al., 2021)	(Dimopoulos & Bakas, Artificial intelligence for mass appraisals of residential properties in Nicosia: Math- ematical modelling and algorithmic implementation, 2019a)	(Nejad et al., 2017)
XGBOOST	0.008	-	0.15	-	-	-	0.039	-	-	0.067
Article Algorithm	(Chanasit et al., 2021))	(Çiğın & Gökçen, 2023)	(Lahmiri et al., 2023)	(Sibindi et al., 2023)	(Mora-Garcia et al., 2022)	(Zaki et al., 2022)	(Chou et al., 2022)	(Kim et al., 2022)	(Abdul-Rah- man et al., 2021)	
ADB	-	-	-	0.588	-	-	-	-	-	
ANN	0.065	0.242	-	-	-	-	3,018,614	64.461	-	
BT	-	-	-	-	-	-	-	-	-	
CAT	-	-	-	-	-	-	-	-	-	
DT	-	-	-	-	-	-	-	-	-	
EN	-	-	-	-	-	-	-	-	-	
ERT	-	-	3.803	-	-	-	-	-	-	
GBM	-	-	-	0.517	0.036	-	-	-	-	
HPM	-	-	-	-	-	-	-	-	-	
KNN	-	-	-	-	-	-	-	-	-	
LASSO	-	0.299	-	-	-	-	-	-	-	
LL	-	-	-	-	-	-	-	-	-	
LR	-	0.299	-	-	0.216	-	3,747,564	-	-	
LRIDGE	-	-	-	-	-	-	-	-	-	

Table 3 (continued)

Article Algorithm	(Chanasit et al., 2021))	(Çulgin & Gökgen, 2023)	(Lahmiri et al., 2023)	(Sibindi et al., 2023)	(Mora-Garcia et al., 2022)	(Zaki et al., 2022)	(Chou et al., 2022)	(Kim et al., 2022)	(Abdul-Rah- man et al., 2021)
MLR	-	-	-	-	-	-	-	-	0.181
RF	-	-	-	-	0.018	-	-	48,189	-
RIDGE	-	0.298	-	-	-	-	-	-	0.195
SVR	-	-	5.141	-	-	-	3,426,277	-	-
XGBOOST	-	0.242	-	0.295	0.051	2.73	-	-	0.148

Table 4 Ratio of the Root Mean Square Error (RMSE) obtained in each study. Source: Author

Article Algorithm	(Almaslukh, 2020)	(Rico-Juan & de La Paz, 2021)	(Castelli et al., 2020)	(Ho et al., 2020)	(Tchuente & Nyawa, 2021)	(Li et al., 2021)	(Peng, 2021)	(Levantesi & Piscopo, 2020)	(Kang et al., 2020)	(Mrsic et al., 2020)
ADB	-	0.354	-	-	92,671	-	-	-	-	180,306
ANN	-	0.206	0.065	-	84,440	-	-	-	-	-
BT	-	-	-	-	-	-	-	-	-	-
CAT	-	0.161	-	-	-	-	-	-	-	-
DT	-	0.31	-	-	-	-	0.829	-	-	252.67
EN	-	-	0.065	-	-	-	-	-	-	-
ERT	-	-	-	-	-	-	-	-	-	-
GBM	0.013	-	-	0.089	91,744	-	-	-	0.077	177,321
HPM	-	-	-	-	-	0.153	-	-	-	-
KNN	-	0.163	0.064	-	86,999	-	-	-	-	253,218
LASSO	-	-	0.064	-	-	-	-	-	-	-
LL	-	0.613	-	-	-	-	-	-	-	-
LR	-	0.259	-	-	97,007	-	-	-	0.103	227,673
LRIDGE	-	0.259	-	-	-	-	-	-	-	-
NLR	-	-	-	-	-	-	-	-	-	-
RF	-	0.077	0.073	0.089	85,259	-	0.726	8505	-	-
RIDGE	-	-	0.065	-	-	-	-	-	-	-
SVR	0.038	-	0.066	0.119	113,420	-	-	-	-	-
XGBOOST	0.013	0.2	-	-	-	0.057	-	-	-	176,500

Table 4 (continued)

Article Algorithm	(Chiramel et al., 2020)	(Dimopoulos & Bakas, 2019)	(Masrom et al., 2019)	(Mohd et al., 2019)	(Nejad et al., 2017)	(Oladunni et al., 2017)	(Guo et al., 2020)	(Chanasit et al., 2021)	(Niu & Niu, 2019)
Sensitivity of machine learning models for the mass appraisal of real estate. Case study of residential units in Nicosia, Cyprus, 2019)									
ADB	-	-	-	-	-	-	-	-	6.72
ANN	-	-	-	-	-	-	-	0.378	-
BT	-	-	-	-	-	-	-	-	-
CAT	-	-	-	-	-	-	-	-	-
DT	-	-	0.056	0.056	-	-	-	-	-
EN	-	-	-	-	-	-	-	-	-
ERT	-	-	-	-	-	-	-	-	-
GBM	-	35.946	-	-	0.092	-	-	-	7.58
HPM	-	-	-	-	-	-	-	-	-
KNN	-	-	-	-	-	-	-	-	-
LASSO	-	-	0.283	0.283	-	-	-	-	-
LL	-	-	-	-	-	-	-	-	-
LR	0.0178	30.048	0.267	0.267	-	-	-	-	-

Table 4 (continued)

Article Algorithm	(Chiramel et al., 2020)	(Dimopoulos & Bakas, 2019)	(Masrom et al., 2019)	(Mohd et al., 2019)	(Nejad et al., 2017)	(Oladunni et al., 2017)	(Guo et al., 2020)	(Chanasit et al., 2021)	(Niu & Niu, 2019)
Sensitivity analysis of machine learning models for the mass appraisal of real estate. Case study of residential units in Nicosia, Cyprus, 2019)									
LRIDGE	-	-	-	-	-	-	-	-	-
NLR	-	29.501	0.045	0.045	-	-	-	-	-
RF	-	27.951	-	-	0.087	96,320	3964.72	-	8.36
RIDGE	0.018	-	0.267	0.267	-	-	-	-	-
SVR	-	-	-	-	-	-	-	-	-
XGBOOST	-	-	-	-	0.099	-	-	-	-

can further support the justification for it being the most adopted algorithm in the reviewed articles, applied in 31 out of the 63 analyzed studies. For instance, Rico-Juan and de La Paz (2021), using the RF algorithm, achieved an R^2 value of 0.986, MAE of 0.03, and RMSE of 0.077%. The authors collected 392,412 instances between 2004 and 2012, using 53 attributes as input.

4 Conclusions

The present work conducted a systematic literature review of open-access articles that employ machine learning (ML) for real estate pricing. This theme is widely disseminated worldwide, with studies found in 21 different countries, although concentrated in Asia (45%). The analysis indicated a growing number of studies in this field over the years.

The size of the dataset collected for implementing the models was highly diverse, ranging from 89 to 12,223,582 instances. The highest frequency was observed in the range of 20,000 to 50,000 samples, with 14 studies (22% of the total). The data sources also exhibited significant variability, with data collected from real estate sales websites (34%), public agency directories (31%), and data repositories in the field of data science (17%). No clear correlation was found between the quality of the model and the size of the dataset; nor with the dataset's regional coverage (municipal, state, or country level).

Studies were identified that utilized both advertised real estate data and data on properties that were actually sold. It was observed that using data from properties that have been sold helps to avoid issues related to price fluctuations that depend on market conditions at the time of data collection. Additionally, false property ads can negatively impact model performance by introducing inaccurate information, which affects the models' predictive capabilities.

A total of 26 main input attributes (features) were identified to predict the selling price, which can be related to the property's location, physical components of the building (infrastructure), and the financial expenses involved in their acquisition, utilization, and maintenance. A few authors investigated which features were the most influential, but they showed disparate results. A direct relationship between the number of attributes used in the models and their evaluation metrics was not detected.

The most adopted ML techniques in this literature review were Random Forest (RF), Extreme Gradient Boosting (XGBoost) + Gradient Boosting Machine GBM, Artificial Neural Network (ANN), Linear Regression and Support Vector Regression (SVR). The most commonly adopted metrics to assess model quality were Root Mean Square Error (RMSE), Coefficient of Determination (R^2), Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE) and Mean Square Error (MSE). Out of 11 authors that compared multiple algorithms including RF, 7 reported RF presenting the best metrics. No other trend was observed between the type of algorithm and the quality of the model, even for datasets of similar sizes. In conclusion, there is no unique technique/algorithm most suitable for predicting property sale prices, as several factors affect metric results.

As knowledge gaps, no investigation was found addressing the relationship between the time between listing and sale with the selling price. This is an interesting demand to represent market trends more realistically, assessing the price ranges that can be applied for a faster property sale. Also, in the vast majority of the articles,

no discussion was found related to the application of a correction index for updating prices, ensuring that the data used for model training does not become outdated to predict future selling prices. The procedures to adjust property values to account for issues of temporality, seasonality, market oscillations, and inflation over the years, when seldom conducted, have not been discussed in depth.

The same can be said for encoding strategies to deal with categoric variables, which are crucial for this type of problem. None of the reviewed works dove into the subject or compared the numerous techniques that can be applied to this purpose. The studies that did address the transformation of categorical variables into numerical ones focused on a single technique. Some employed a binary treatment, while others only mentioned that categorical variables were transformed but did not specify the process.

In general, a severe lack of clarity is evident in several aspects of the articles' methodologies, including the description of data processing procedures, the information collecting criteria, treating of non-numeric variables, and model building, making most studies irreproducible. Lack of clarity in data and treatments can lead to unreliable ML models, making it difficult for decision-makers to interpret and justify the models' decisions, which is critical in industries that require transparency. This also makes accountability for decisions problematic and can undermine public trust, limiting acceptance of AI-based technologies in the real state sector.

In conclusion, the studied ML techniques have proven to be suitable for predicting property prices, provided that they are fed by representative databases, the models are constructed ethically, and they are evaluated using relevant metrics. No single superior ML technique was found, as their performance depends on various database characteristics. Therefore, it is recommended that researchers experiment with various techniques. This systematic review aims to contribute to the improvement and evolution of studies related to ML in the real estate market, with the goal of enhancing the reliability and accuracy of results.

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Declarations

Competing interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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